

23. Prompt engineering is the practice of designing, structuring and refining the instructions (or "prompts") given to artificial intelligence models to get the most accurate, useful and relevant output.

It is important because the quality of a AI's output depends directly on how clearly and strategically you communicate your intent through your input. It improves accuracy and quality, saves time and boosts efficiency.

Poor written Prompts.

- * Write something good for our website.
- * Write an article about social media.

Corrected version Prompts.

- * Create a 300 word 'About Us' section for our IT consulting website. Highlight our 15 years of experience helping mid market healthcare clients and emphasize our security certification. Use a professional and reassuring tone that avoids heavy technical jargon.
- * Write a 500 word article about the impact of social media on teenagers' mental health. Use a conversational tone, include three practical tips for parents, and format the tips as bullet point.

24. A map function is a foundational concept in computer science where a specified transformation function is applied to every element of a data collection (such as list, array, or tuple).

It is useful because it makes code shorter and cleaner. It helps apply consistent changes across large datasets.